

Aditya Kulkarni

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PROFESSIONAL SUMMARY

MS Computer Science candidate (NC State, May 2026) with production experience building full-stack TypeScript and React applications used by real users. Shipped 10+ backend features at Isomer AI across 7 production deployments (394 commits) -- owning requirements, implementation, testing, and monitoring end-to-end. Built real-time systems including Server-Sent Event leaderboard feeds, Redis-backed live updates, and Pusher event pipelines. Strong across the full stack: TypeScript/React/Next.js frontend, Node.js/PostgreSQL/Redis backend, Auth0 RBAC, REST APIs, and Docker CI/CD. Fast product thinker -- 3 complete MVP builds in 48-hour hackathon builds, including HackNC24 Top 10 of 100+ teams.

CORE COMPETENCIES

TypeScript + React + Next.js

Node.js + REST API

PostgreSQL + Redis

Real-Time Systems (SSE + WebSockets)

Auth0 + OAuth + RBAC

Full-Stack Product Delivery

Docker + CI/CD + GitHub Actions

Python + AI/ML Pipelines

WORK EXPERIENCE

Isomer AI May 2025 - Present

Product Engineer Intern | Remote

- Shipped **10+ full-stack features** end-to-end for a production AI-powered claims platform -- TypeScript, Node.js, Express, PostgreSQL, Redis, Prisma, Pusher -- across 7 deployments with 394 total commits; owned requirements through monitoring for each feature.
- Implemented **Auth0 RBAC** with multi-provider token validation, real-time event pipelines (Pusher + SSE), document processing (PDF generation, Microsoft Graph API email parsing, S3), and multi-model AI agent tooling across 8+ models (Claude, GPT-4o, Gemini) with MCP and Temporal.io orchestration.

NC State University April 2025 - Present

Graduate Research Assistant | Raleigh, NC

- Designed a cloud-based genomic data platform (JBrowse 2 + Google Cloud Run) serving 500GB interactive datasets to 100+ researchers worldwide; collaborated on ML architecture improving model accuracy by 12%.

Mphasis (HP Company) Jun 2024 - Aug 2024

Software Developer Intern | Remote

- Built 5+ AI-driven automation workflows (OCR, NLP, ML classification) improving claim accuracy by 25%; automated deployments on AWS and GCP with Grafana monitoring and reliability testing.

Capgemini Jun 2023 - Aug 2023

AI/ML Intern | Remote

- Built distributed data pipelines with Apache Kafka and Spark for real-time streaming and batch processing, reducing ticket-processing latency by 20% and manual processing time by 15%.

PROJECTS

PulseRank -- Real-Time Leaderboard & Activity Feed Real-Time / Full Stack

Event-driven leaderboard and activity feed platform built with real-time collaboration in mind: MySQL for transactional event writes with idempotency-key deduplication and outbox pattern, MongoDB for TTL-expiring feed projections, **Redis sorted sets for sub-millisecond leaderboard reads**, and **live score updates over Server-Sent Events**. Validated multi-store consistency with Testcontainers; benchmarked hot paths with k6; shipped observability with OpenTelemetry + Prometheus and GitHub Actions CI/CD.

TypeScript · NestJS · Next.js · MySQL · MongoDB · Redis · Docker · OpenTelemetry · Prometheus | github.com/Aditya-k24/PulseRank

AlgoPulse -- Agentic AI Platform 0-to-1 Product Build

Architected an event-driven AI orchestration platform from scratch. Identified an API-polling bottleneck under load and redesigned to **Supabase Edge Functions streaming AI reasoning loops directly to the client**. Scaled to 2,000 async inference requests/second with Temporal.io for durable workflow state management and Kafka for fault-tolerant synchronization -- zero state loss under load.

TypeScript · Node.js · React · Supabase Edge Functions · Temporal.io · Kafka | github.com/Aditya-k24/AlgoPulse

VectorLift -- Semantic Search + Ranking Engine Python / Full Stack

Production-grade search system on MS MARCO: BM25 (Elasticsearch), dense bi-encoder retrieval (PyTorch fine-tuned), and cross-encoder re-ranker. Kafka-driven indexing pipeline; **FastAPI serving API** with per-stage latency tracking; Streamlit dashboard for experiment comparisons. Full stack deployed via Docker Compose with PostgreSQL, Redis, Prometheus, and Grafana. Evaluated 6 pipeline configurations with NDCG@10, MRR@10, MAP, and paired bootstrap significance testing.

Python · PyTorch · FastAPI · PostgreSQL · Redis · Kafka · Elasticsearch · Qdrant · Streamlit · Docker | github.com/Aditya-k24/VectorLift

PerCent -- GenAI Finance App HackNC24 Top 10

Full-stack GenAI finance app built in 48 hours: goal tracking, expense logging, and a RAG-based AI chatbot (Selenium scraping + Hugging Face + Qwen-14B-Chat) reducing manual financial research by 30%. Ranked Top 10 of 100+ teams at HackNC24.

Python · React · Hugging Face · Qwen-14B · RAG · Selenium | github.com/adityapai18/HackNC24

EDUCATION

MS Computer Science -- NC State University Aug 2024 - May 2026

Courses: Artificial Intelligence, Operating Systems, Data Science, Object Oriented Development, Software Engineering, Algorithms

BTech Computer Science -- Mumbai University Aug 2020 - May 2024

CGPA: 9.32/10 | Courses: DS & Algorithms, OS, Machine Learning, Software Engineering, Data Warehousing, Big Data Infrastructure

SKILLS

Frontend: TypeScript, React, Next.js, Tailwind CSS, Streamlit

Backend: Node.js, NestJS, Express, FastAPI, Spring Boot, REST API, GraphQL, Temporal.io

Databases: PostgreSQL, MySQL, MongoDB, Redis, Elasticsearch, Qdrant **Auth/Security:** Auth0, OAuth 2.0, JWT, RBAC, Spring Security

Languages: TypeScript, Python, Java, JavaScript, SQL, C, R

DevOps/Cloud: Docker, Kubernetes, GitHub Actions, AWS, GCP, Prometheus, Grafana, OpenTelemetry